PRODUCTTANK BENGALURU

Ahead of the Scale.

Two thousand three hundred years of road building, and the new driver who arrived on a road somebody else designed.

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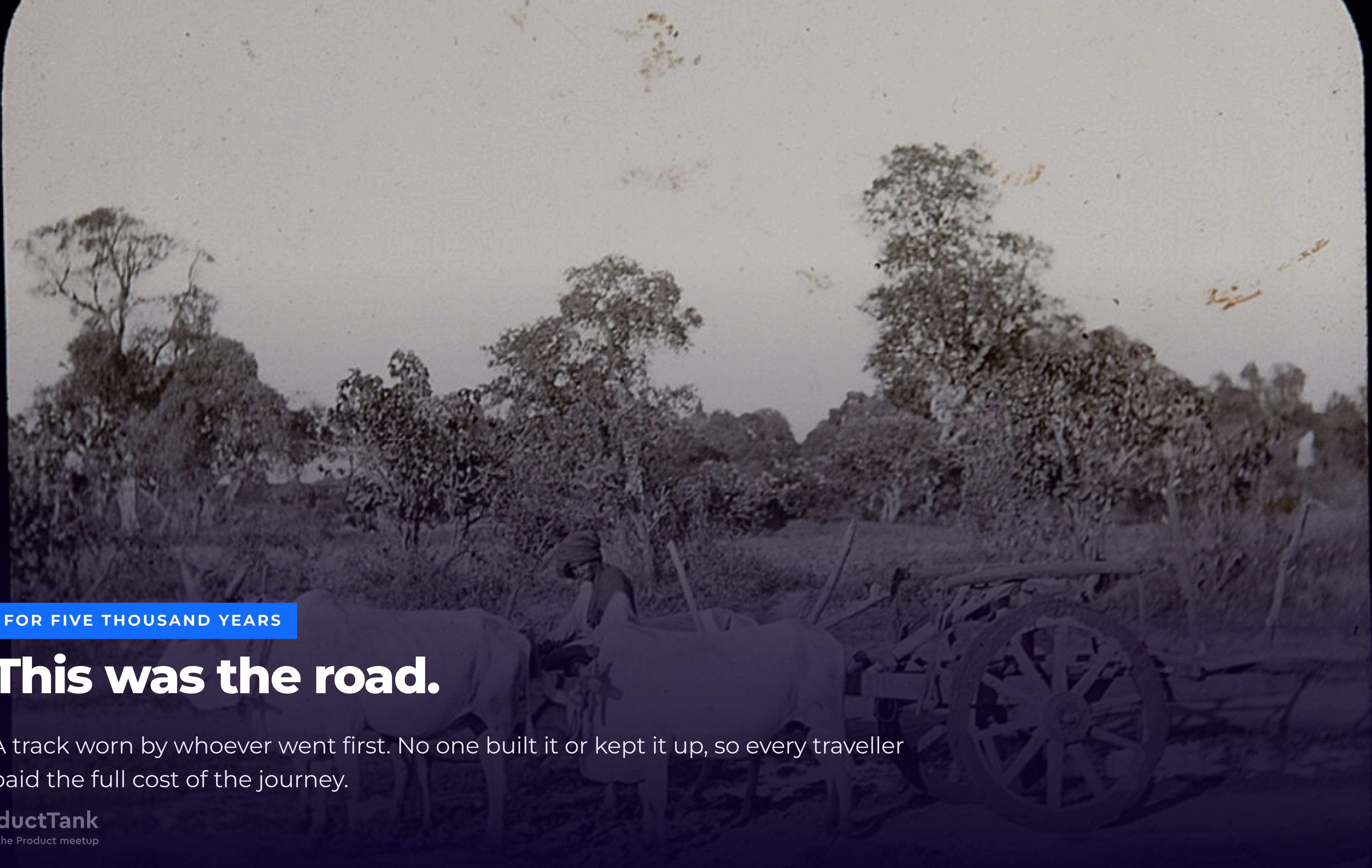


FOR FIVE THOUSAND YEARS

This was the road.

A track worn by whoever went first. No one built it or kept it up, so every traveller paid the full cost of the journey.

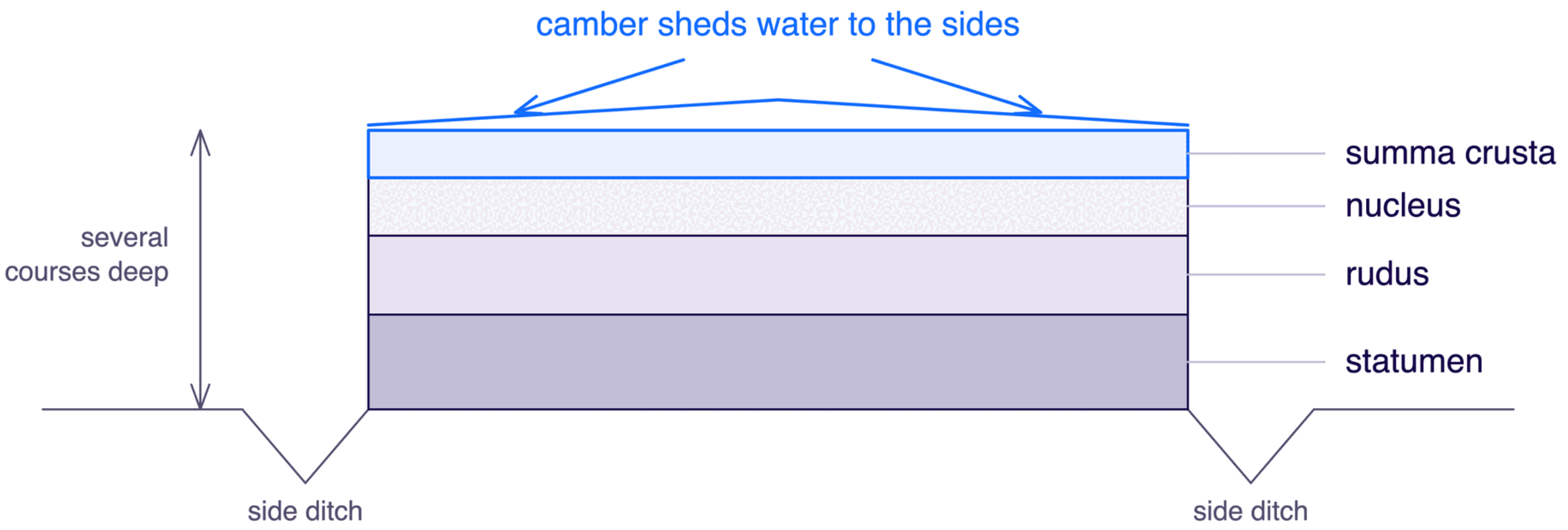
312 BC

Then somebody built one on purpose.

And the difference lasted two thousand years.

The Appian Way

Construction of the **Via Appia** began in **312 BC**. What the Romans chose to worry about is the interesting part:



A third thing this drawing cannot show: **connected routes** — a network joined to other networks.

THE ROAD

Once somebody builds a road properly, every journey after it becomes cheaper — because the decisions are made once, in the road itself, instead of again by every traveller.

540 KILOMETRES

Rome to Brundisium.

Roughly Bengaluru to Goa — laid by hand, to a fixed spec. **White:** the Via Appia, from 312 BC. **Pink:** the Via Traiana, Trajan's faster alternative four centuries on.

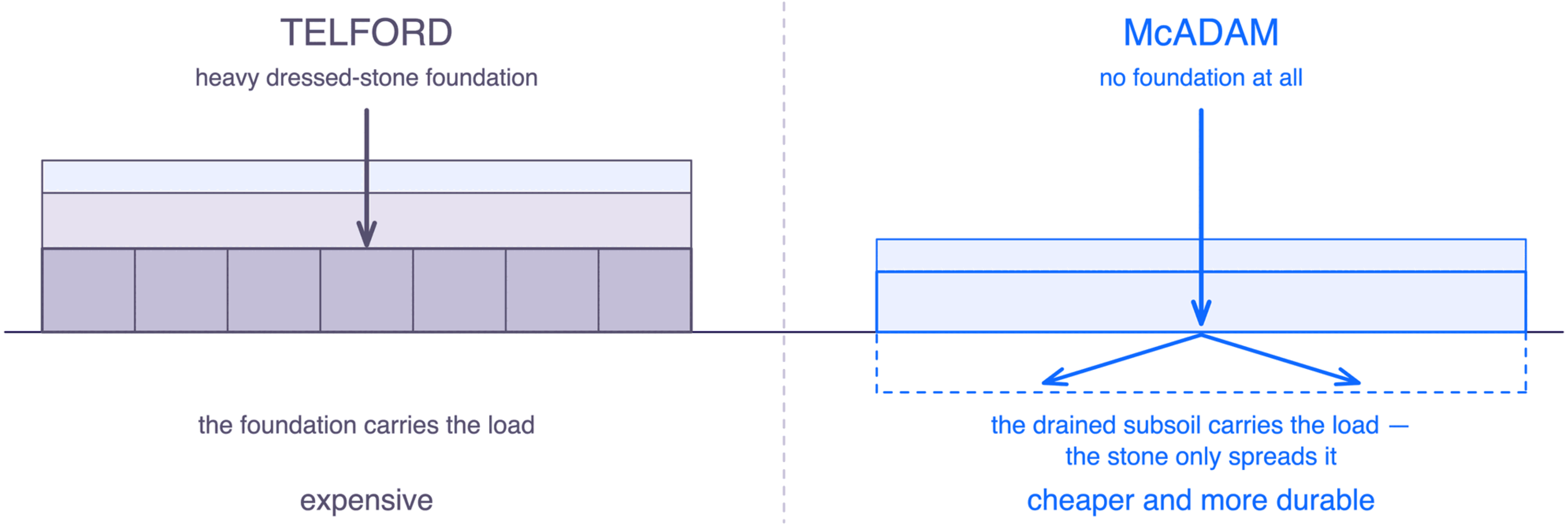
1800S

Two thousand years later.

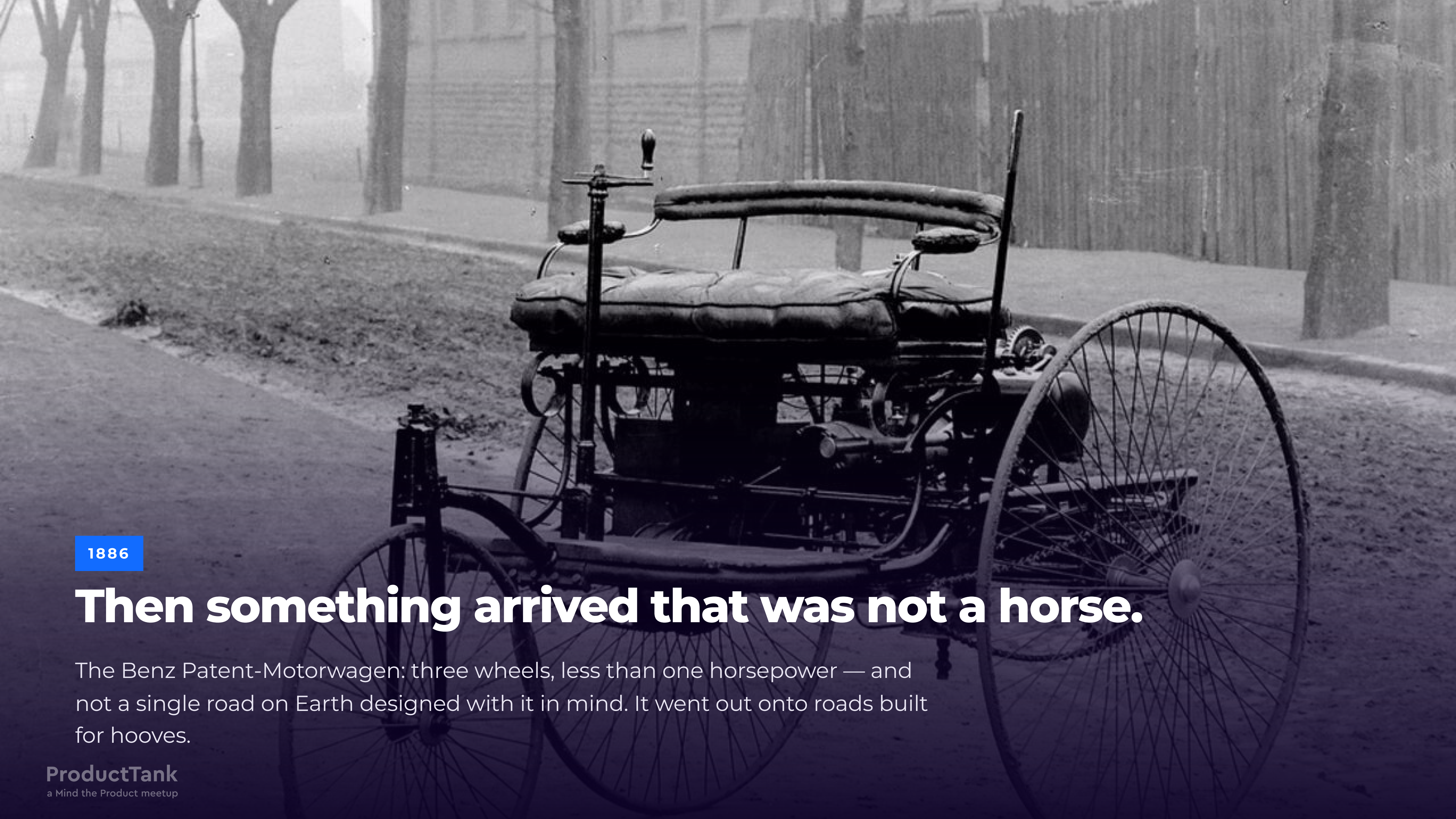
The fastest way to cross a country was still a horse.

Macadam

John Loudon McAdam worked out how to build a road that was **cheaper and more durable** at the same time: compacted layers of graded broken stone, sized to a gauge, drained, and **without** the expensive heavy stone foundation his contemporaries insisted on.



The method spread through Britain and well beyond it, because anyone could afford it. Better and better roads — built, every one of them, for the horse.



1886

Then something arrived that was not a horse.

The Benz Patent-Motorwagen: three wheels, less than one horsepower — and not a single road on Earth designed with it in mind. It went out onto roads built for hooves.

AUGUST 1888

Then somebody proved it was useful.

Bertha Benz

In **August 1888**, **Bertha Benz** drove from **Mannheim to Pforzheim**, roughly **106 km one way**, with her two sons, and without telling her husband.

She dealt with the problems as they arrived: a blocked fuel line, worn brakes, failing insulation. She fixed them herself, at the roadside, with what she had.

She proved something no patent could: the machine **worked over real distance, in the real world**.

Mannheim → Pforzheim, ~106 km one way, August 1888.

1896

On roads built for somebody else.

For the next twenty years the machine kept getting faster. The road did not change.

The man with the red flag

Britain's **Locomotives Act 1865** required a person to walk **at least 60 yards ahead** of certain road locomotives, **carrying a red flag**.

60 yd	4 / 2 mph	1
ESCORT DISTANCE AHEAD	COUNTRY / TOWN LIMIT	HUMAN PER MACHINE

Picture a powerful new machine with a person walking in front of it, waving a flag.

1913

Then it got cheap.

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a Mind the Product meetup

The scale curve

Registered motor vehicles in the United States:

8,000	~10M	26.7M
1900	1920	1930

Eight thousand to twenty-six point seven million, in thirty years.

Somewhere in the middle of that curve, everybody sorting it out for themselves stops working. Nobody did anything wrong. The cost of everybody agreeing separately grows faster than the traffic does.

Source: FHWA Highway Statistics historical series (MV-200).

THE 1900S

This was the coordination mechanism.

One man, one junction, making every decision fresh for as long as he stood there.

THE 1920S

And everything broke.

Eleven signs for one route

Early automobile clubs put up their own route signs. Each club had its own scheme, and none of them coordinated.

The Federal Highway Administration's own history records that on some heavily travelled routes, a driver could encounter **as many as eleven different signs for a single route**.

Eleven signs. One road. **Every one of them technically correct.**

Source: FHWA highway history.

A line painted on a road

The fix was already on the road. In **1911** — a decade before that traffic jam — somebody had painted a **centreline** down the middle of a road in Michigan. The idea is credited to **Edward N. Hines**.

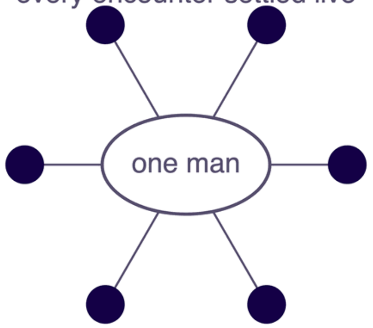
It is a bucket of paint.

WHAT THE LINE DOES

Two strangers moving toward each other at speed coordinate without ever meeting, negotiating, or learning anything about each other.

BEFORE

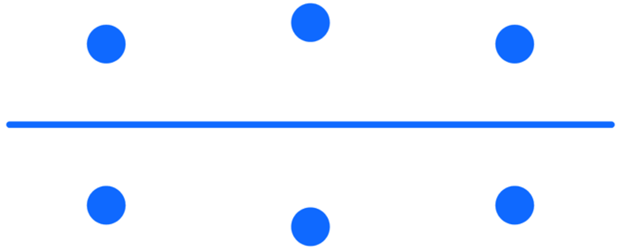
every encounter settled live



decided fresh, one at a time —
and only while he is standing there

AFTER 1911

the decision lives in the road



nobody negotiates anything —
the line decided once, for everyone

Then, very quickly

1911

Centreline

Painted road marking, Michigan.

1914 → 1920

Traffic signals

First permanent electric signal, Cleveland. Then the tri-colour signal, Detroit — credited to William Potts.

1923

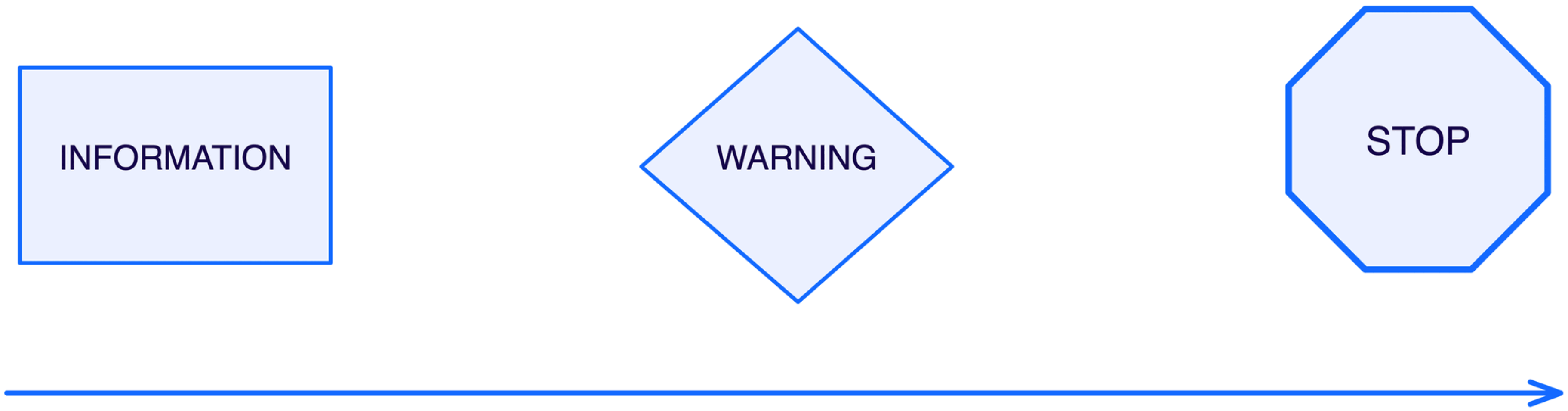
Shape = meaning

Sign shapes standardised, so meaning is readable before the text is.

In roughly a decade, the road went from carrying **no information** to carrying instructions a stranger could read at speed, in the dark, in a language they did not speak.

Shape carries the meaning

In **1923**, the **Mississippi Valley Association of State Highway Departments** proposed something clever: encode the **severity** of a sign in its **shape**, so the meaning survives darkness, distance, a foreign language, and a driver who has not read the text yet.



the more sides, the more dangerous

The octagon sits at the top of that hierarchy, and it is still on every street in this city.

Mississippi Valley Association of State Highway Departments, 1923. AASHO's combining report followed in 1924; its manual in 1927.

Convergence

Two bodies published separate manuals, one for rural roads, one for urban streets. Two sets of rules, both reasonable, and inconsistent with each other.

It took a **third effort to unify them**: the first **Manual on Uniform Traffic Control Devices**, in **1935**.

> *"Uniformity of devices simplifies the task of the road user because it aids in recognition and understanding, thereby reducing perception/reaction time."*
> *— MUTCD §1A.06*

Read the justification again. The stated purpose is **cutting the time it takes somebody to understand what is happening.**

First MUTCD, 1935. The quotation is MUTCD §1A.06, verbatim.

1903 ONWARD

Roads do not drive themselves

Britain introduced **driver licences in 1903**, largely for **identification** rather than competence. You did not have to demonstrate that you could drive.

Requirements to be any good at it came **later**, over decades.

The order was: **build the road first. Encode the rules second. Ask the driver to be competent third.**

Motor Car Act 1903 — licences for identification, not competence.

1930S

Then we rebuilt the road around the machine.

Grade separation. Controlled access. Curves designed for a speed nobody could have reached in 1886.

1956

And did it at continental scale.

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The Interstate programme

The **Federal-Aid Highway Act of 1956** went further than paying for asphalt: Congress created the **mandate** for **uniform geometric and construction standards**, and **AASHO** — the engineering body — wrote the **specification** the federal highway authority adopted. Neither could have done the other's job.

- Controlled access, with grade-separated intersections
- Standardised lane widths
- Design speeds appropriate to terrain

1956, DESIGNING FOR 1975

The standards specified a capacity horizon: engineers sized the roads for the traffic expected in 1975 — traffic that would not exist for twenty years. And they were right.

Federal-Aid Highway Act of 1956; AASHO design standards; FHWA on the twenty-year design period.

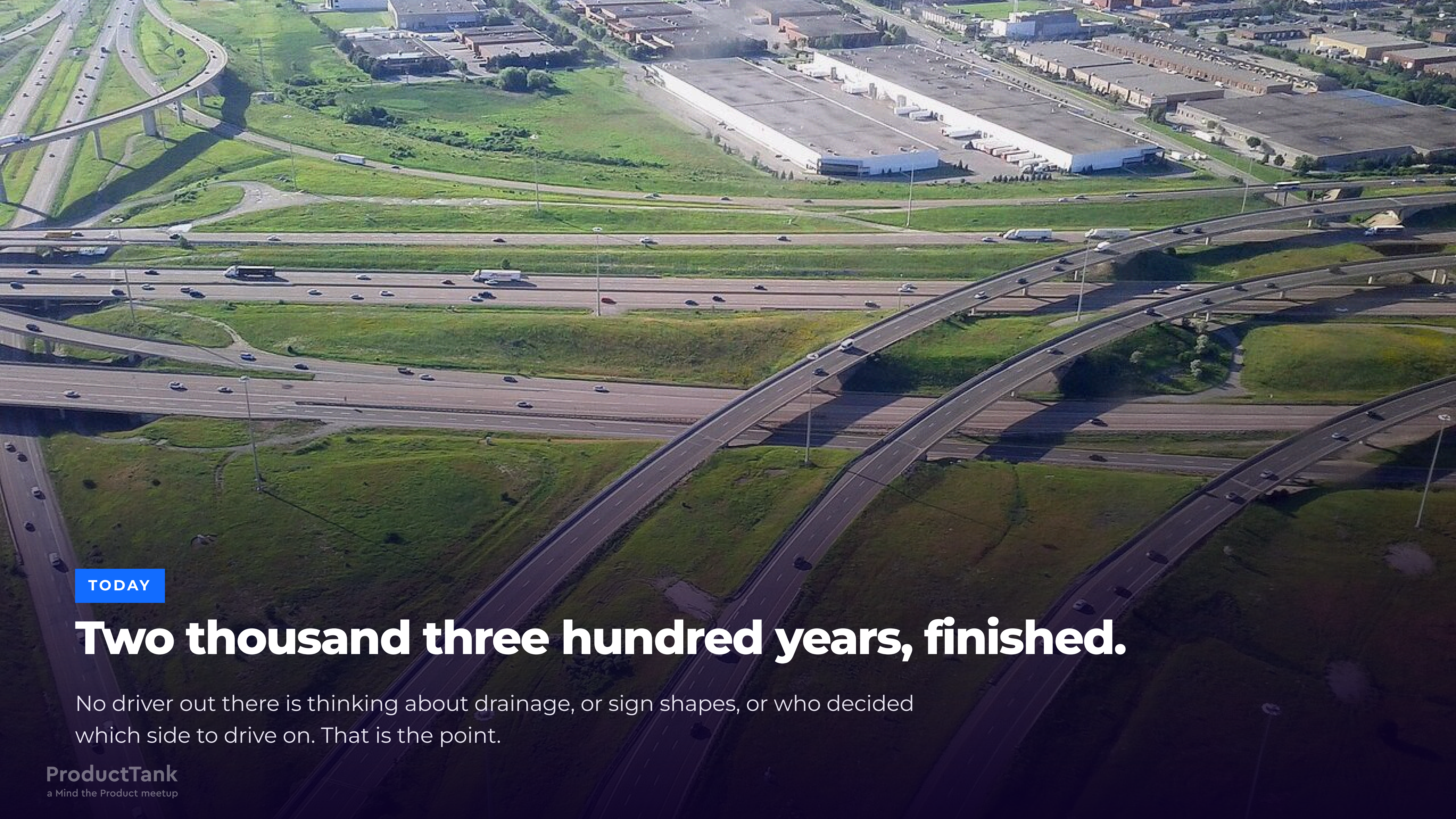
And it worked

Interstates carry traffic far faster than the roads they replaced. If rules and barriers were a tax on speed, you would expect them to be more dangerous.

0.85	1.53	1.8×
INTERSTATE FATALITIES PER 100M VEHICLE-MILES	OTHER ROADS, SAME MEASURE	THE RISK YOU AVOID BY DRIVING THE BUILT ROAD

Faster roads. Roughly half the fatality rate. The guardrails made the speed survivable.

Source: FHWA. The 1.8× is the ratio of the two published FHWA figures.

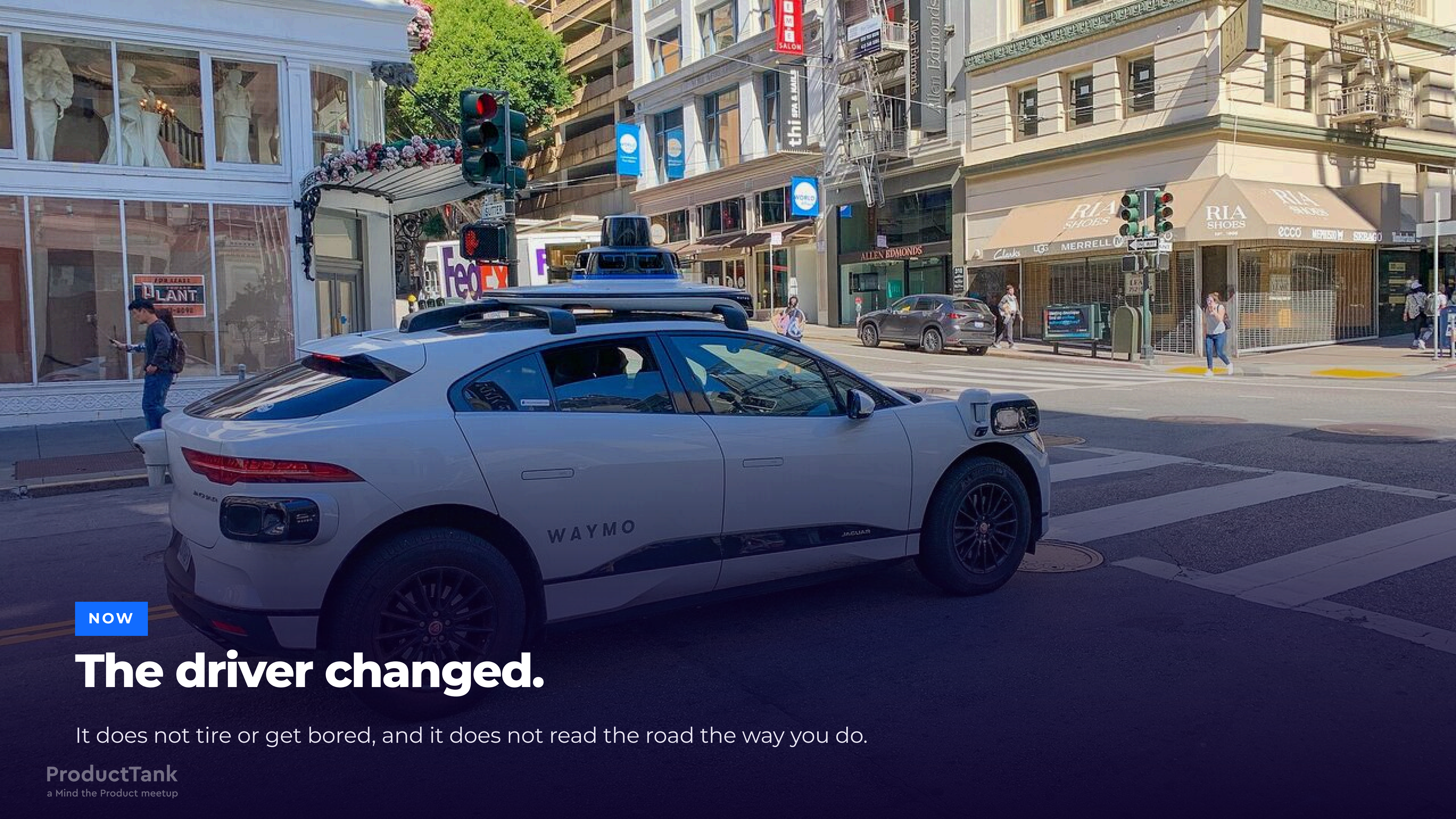


TODAY

Two thousand three hundred years, finished.

No driver out there is thinking about drainage, or sign shapes, or who decided which side to drive on. That is the point.

What happens when the driver changes?



NOW

The driver changed.

It does not tire or get bored, and it does not read the road the way you do.

For over a century, we optimised around one assumption

TRANSPORTATION

Signage legibility, reaction-time margins, lane widths, signal timing, licensing: engineers calibrated each one around the capabilities and limits of a **human driver**.

PRODUCT AND SOFTWARE ENGINEERING

Spec review, code review, roadmap cadence, approval gates, onboarding docs: we calibrated each one around the capabilities and limits of a **human team** — people who decide at meeting speed and ship at review speed.

Both platforms are **human-scaled**. We shaped them to the operator.

And then a new participant arrived

AI agents are now writing, reviewing, deploying and operating software — and increasingly, drafting the specs and the tickets that describe it. They are:

FASTER, AND NEVER OFF

They propose changes faster than you can review them, with no sprint boundary and no Friday afternoon.

HORIZONTALLY SCALABLE

Ten agents cost roughly what one costs.

PROBABILISTIC

The same input need not produce the same output.

CAPABLE OF ACTING

Real writes to real systems.

OCCASIONALLY CONFIDENTLY WRONG

They fail in plausible, well-formatted prose.

This already happened once.

In **1886**, a far more capable participant arrived on infrastructure designed for the previous generation of participant.

The infrastructure was **correctly designed for the wrong user**.

Five assumptions about the driver

Our platform is human-scaled — the process as much as the pipeline. Each of these gates assumes the operator is a person.

Spec review and code review assume a human bottleneck is acceptable. We made the slowest step slow on purpose.

Quarterly roadmaps assume change arrives at human frequency.

Estimates assume throughput is a function of how many people you have.

Approval gates and staging environments assume somebody is watching when the change lands.

Documentation and "ask Amit" assume a reader who asks a colleague when it is unclear, and that the participant can, in fact, ask Amit.

Each of these made sense when we built it. **All of them are calibrated for the previous driver.**

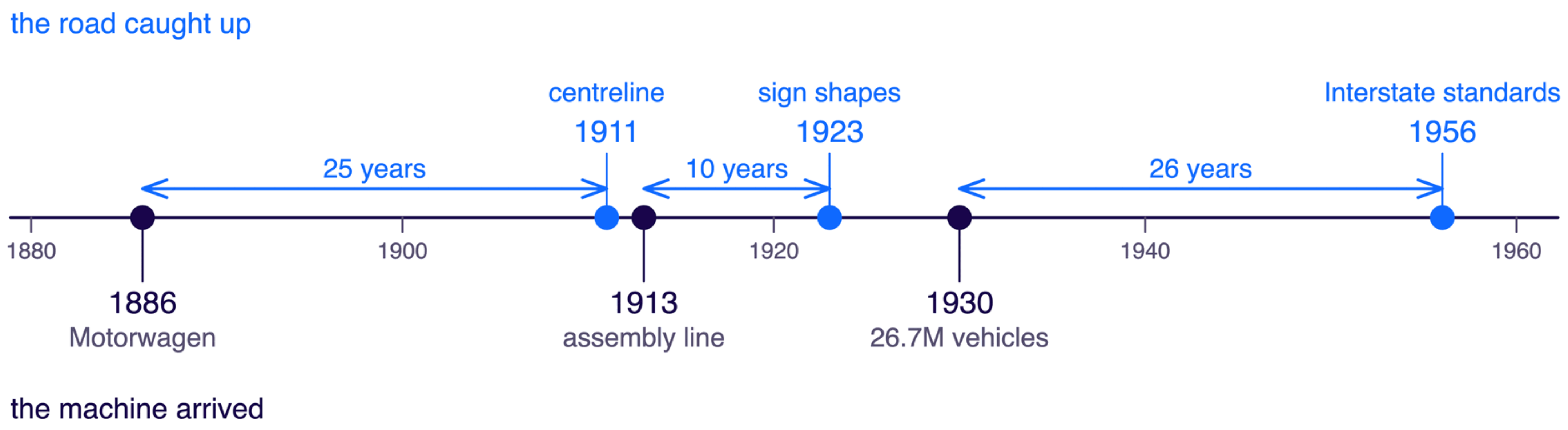
When the driver becomes less predictable, the road must become more predictable.

Meetings will not contain a probabilistic operator. Move the decisions **out** of human review and **into** the environment, where they stay deterministic, checkable and always on.

The obvious conclusion is wrong

If the road has to become more predictable, the tidy conclusion is that you build the road **first**.

That is not what you just watched.



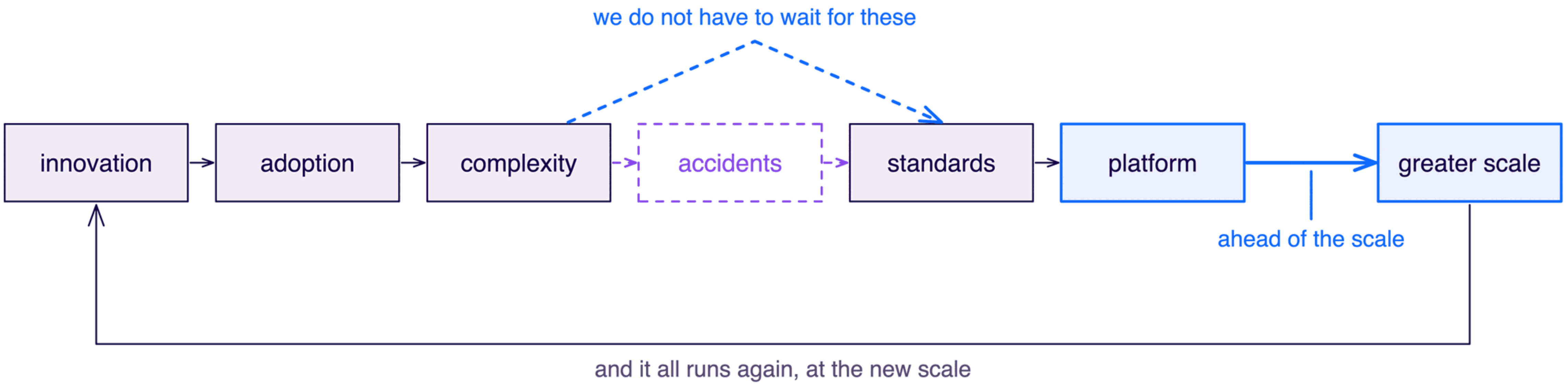
The machine arrived first every time, by decades.

Ahead of the scale

THE THESIS

The platform does not have to be ahead of the innovation. It has to be ahead of the scale.

The cycle the road record documents:



Innovation arriving first is normal and fine. Arriving at **26 million vehicles** still running 1900's coordination model is the failure.

For AI agents, we are somewhere around 1905.

The participant has arrived and adoption is compounding, while the coordination model is still the previous generation's.

We know what comes next in this sequence. **We do not have to wait for the accidents to write the standards.**

THE PART THAT IS MINE

The part where I stop talking about roads

Everything so far was somebody else's history. This next part is mine.

A fifteen-year-old product, and an uncomfortable diagnosis

Homecare: care delivered in the patient's home — nursing, therapy, personal care, remote monitoring — by clinicians in the field, on phones, with unreliable connectivity. Every record is **PHI**, so **there is no "just ship it" path**.

I was leading engineering for the product. Four things were degrading at once — **stability**, **security**, **compliance**, **delivery velocity**. Finance saw four bids for budget; the roadmap saw every estimate quietly growing. They were one problem wearing four costumes.

NO SEPARATION OF CONCERNS

Domain logic, persistence, compliance behaviour and UI reasoning interleaved. You could not change one without auditing all four.

NO PLATFORM FOUNDATION

Every team re-solved identity, audit trails, permissions, forms and data shape, each in its own way.

THE SYSTEM OF RECORD WAS OURS

A bespoke schema meant every new customer concept became a modelling project before it became a feature.

THE DIAGNOSIS

Stability, security, compliance and velocity degrade together, because they share one root cause: the absence of boundaries.

The bet: stop paying to invent our own system of record

FHIR, HL7's Fast Healthcare Interoperability Resources, gave us a schema we could have written ourselves and an ecosystem we could not have.

THE METADATA ALREADY EXISTS

Patient, Encounter, Observation, CarePlan, Practitioner. It already modelled every healthcare concept our customers could name.

FORMS COME WITH IT

Questionnaire and QuestionnaireResponse, plus the Structured Data Capture guide. LOINC supplies the standardised instruments.

YOU NEED NOT HOST IT

Managed FHIR backends exist on all three hyperscalers. Mature self-hosted open source exists too, if you would rather own it.

We stopped paving our own road and drove on somebody else's — one a global community already maintains, tests and extends. That was a build-or-buy decision before it was an architecture decision: not **is the standard perfect**, but **is our own schema worth what we pay, every release, to keep it alive**.

Managed: Azure Health Data Services, AWS HealthLake, Google Cloud Healthcare API. Open source: HAPI FHIR, Medplum, LinuxForHealth FHIR Server.

The pilot: three months, five fresh graduates

We skipped the strategy document and built a narrower version of our own product.

Month 1

Foundations

ABP.io for the application platform: modular, multi-tenant, with audit logging and permission management built in. FHIR for the system of record.

Month 2

A working lite product

A working, narrower version of what we sold.

Month 3

Enterprise-grade, audit-ready

Delivered by five engineers who had graduated that year.

Five graduates. Three months. Audit-ready. Who achieved it mattered more than how fast. If the platform makes new graduates dangerous in a good way, the platform is real — and your delivery estimates stop depending on which five people happen to be free.

The part that made it work

The technology was the easy half. The other half decided the outcome.

01

WE PUT THEM ON THE FLOOR

The team sat in the middle of the floor, not in a side room. They built a startup atmosphere there, and people walking past caught it.

02

WE SOCIALISED IT CONSTANTLY

Openly on the floor, and one-to-one. We pulled people into the discussion as **advisors** — including the stakeholders who could have blocked it — asking for their help and their opinions rather than their sign-off.

03

THEY MADE IT THEIRS

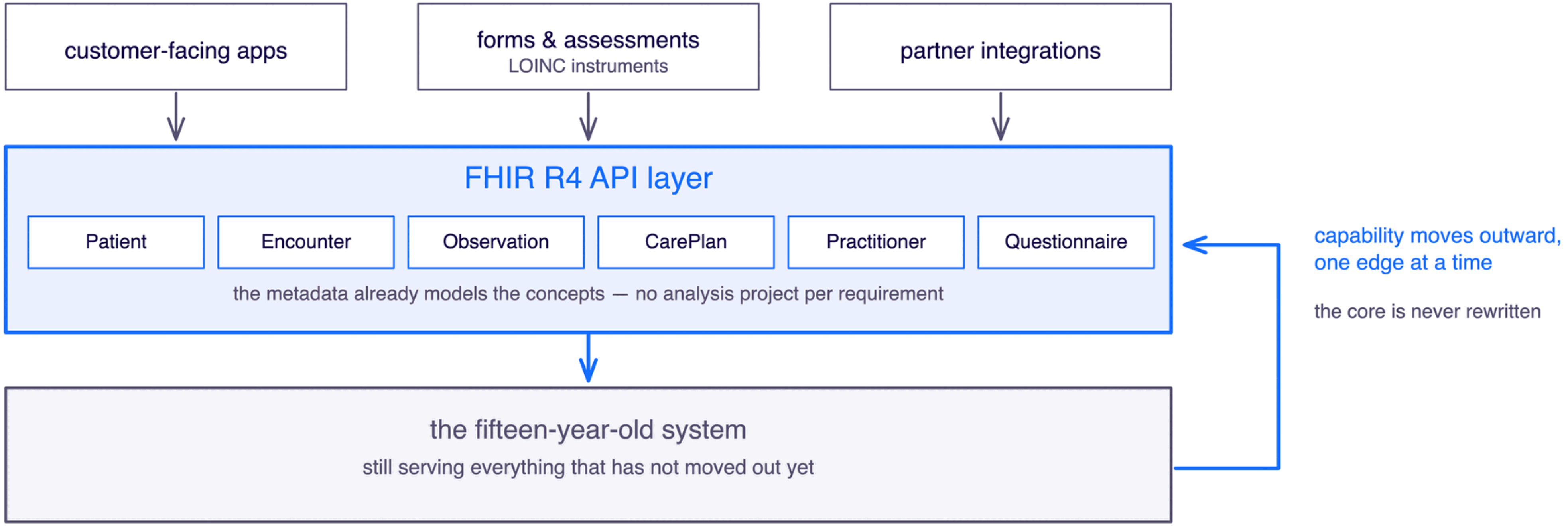
By the end of the three months everybody was contributing and defending it. The vision that had looked impossible now looked reachable.

ADOPTION

People adopt the platform they were consulted on. Ask for advice long before you ask for adoption.

Then we did it to the real product

The pilot was greenfield. The fifteen-year-old system was not — so instead of the big-bang rewrite you keep being asked to fund, we put a **FHIR R4 API layer in front of it**: a strangler fig.



Within **two months**: business analysis stopped being a project, forms stopped being a backlog, and the sceptics converted themselves — the second pilot ran inside the product they already worked on every day.

Strangler fig application, Martin Fowler. The pattern replaces a legacy system incrementally, at the edges, without a big-bang rewrite.

We stopped re-deciding problems that were already solved.

That is where the speed came from. Five graduates proved it before we asked anyone to believe it.

CLOSE

What to do on Monday

Five questions for your platform — and your process

1. COUNT YOUR SIGNS

How many ways can a decision get made here, and how many ways can a change reach production? If either answer is more than two, you have the 1920s problem.

2. FIND THE RED FLAGS

Which of your gates — approvals, sign-offs, review meetings — exist because a human used to be the slowest part? Is that still true?

3. NAME YOUR CAPACITY HORIZON

1956 designed for 1975. What load is your product built for, and do your roadmap and your platform name the same number?

4. FIND YOUR 106 KM

Bertha Benz settled the argument by driving the thing, not by describing it. What is your existence proof, and who is refusing to fund you until they see one?

5. PAINT ONE LINE

Pick the single decision your org re-negotiates most — the one argued in every planning meeting — and encode it once, in the environment, this quarter.

The five takeaways

THE ROAD

Once somebody builds a road properly, every journey after it becomes cheaper — because the decisions are made once, in the road itself, instead of again by every traveller.

THE DRIVER

When the driver becomes less predictable, the road must become more predictable.

THE THESIS

The platform does not have to be ahead of the innovation. It has to be ahead of the scale.

THE DIAGNOSIS

Stability, security, compliance and velocity degrade together, because they share one root cause: the absence of boundaries.

ADOPTION

People adopt the platform they were consulted on. Ask for advice long before you ask for adoption.

Faster roads. Half the fatality rate.

The guardrails made sustained speed possible.

Your agents are the fastest participant your platform has ever had. **Go build them a road.**

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Thank you

Questions, arguments and corrections all welcome, especially the corrections. I had to make nine of them to this talk already.

Slides **github.com/v1bh0r/pt-bengaluru-talk-aug-2026**

Sources Every claim is dated and sourced — **full bibliography**

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